

**CLASSROOM CONTACT PROGRAMME**

(Academic Session : 2020 - 2021)

JEE(Advanced)**REVIEW TEST****27-12-2020****JEE(Main + Advanced) : NURTURE COURSE [PHASE : S, I(A), I, II, III, IV(A) & IV]****PAPER-1****Time : 3 Hours****Maximum Marks : 186****READ THE INSTRUCTIONS CAREFULLY****GENERAL :**

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR**Please see the last page of this booklet for rest of the instructions**

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16,
Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24,
Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127,
Xe = 131, Ba = 137, Ce = 140, Cu = 63.5, Ne = 20, K = 39, Mn = 55

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

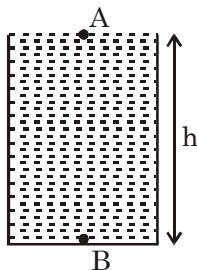
BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I(i) : (Maximum Marks : 12)

- This section contains **FOUR** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

1. Breaking stress of a material of density ρ is σ_0 . Find maximum length of rod of uniform cross-section of this material which can be hanged vertically without breaking :
 (A) $\frac{\sigma_0}{\rho g}$ (B) $\frac{\sigma_0}{2\rho g}$ (C) $\frac{2\sigma_0}{\rho g}$ (D) None
2. A container of height h fully filled with water of density ρ is falling with acceleration $\frac{g}{2}$. A is top most point while point B is bottom most point of tank then pressure difference between point A & B is: (P_0 is atmospheric pressure)



- (A) ρgh (B) $P_0 + \frac{\rho gh}{2}$ (C) $\rho gh - P_0$ (D) $\frac{\rho gh}{2}$

Space for Rough Work

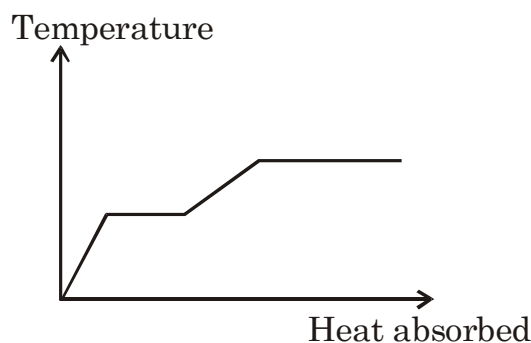
3. A monoatomic ideal gas is compressed isothermally at temperature 300 K from volume V_1 and V_2 then again expanded adiabatically to final volume $2\sqrt{2} V_2$. Find final temperature of gas.
- (A) 150 K (B) $\frac{150}{\sqrt{2}}$ K (C) $600\sqrt{2}$ K (D) None
4. Temperature and volume of a monoatomic ideal gas are related as $T \propto \frac{1}{V}$ in a process. Then molar specific heat capacity of gas in the process is :
- (A) $\frac{3R}{2}$ (B) $\frac{5R}{2}$ (C) $\frac{R}{2}$ (D) 2R

Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 32)

- This section contains **EIGHT** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -1 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

5. Heat is supplied to a certain homogeneous solid sample of matter at a uniform rate. Heat absorbed and temperature is plotted as shown in figure. Which of the following conclusion can be drawn?



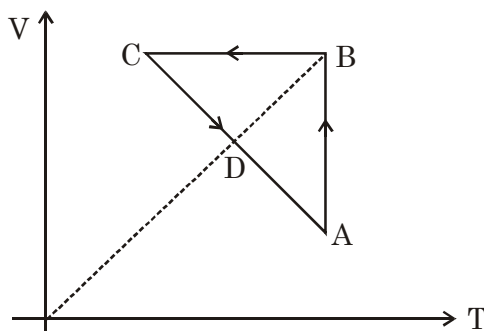
- (A) Its specific heat capacity is greater in solid state than in liquid state.
- (B) Its specific heat capacity is greater in liquid state than in solid state.
- (C) Its latent heat of vaporization is greater than latent heat of fusion.
- (D) Its latent heat of fusion is greater than latent heat of vaporization.

Space for Rough Work

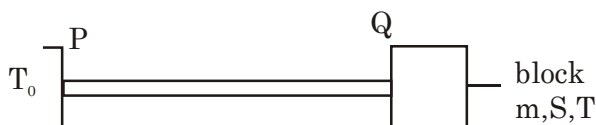
6. A black body emits radiation at the rate P when its temperature is T . At this temperature, the wavelength at which radiation has maximum intensity, is λ_0 . If at temperature T' power radiated is P' and wavelength at maximum intensity is $\frac{\lambda_0}{3}$, then :-
- (A) $T' = 3T$ (B) $T' = \frac{T}{3}$ (C) $P' = 81 P$ (D) $P' = \frac{P}{81}$
7. Two vessels of the same volume contain the same ideal gas at same temperature. If the pressure in the vessels be in the ratio of 1 : 2, choose the incorrect option (s) :-
- (A) The ratio of the average kinetic energy per molecule is 1 : 2
 (B) The ratio of the root mean square speed is 1 : 1
 (C) The ratio of the average speed is 1 : 2
 (D) The ratio of number of molecules is 1 : 2
8. n mole of monoatomic ideal gas is taken from temperature T_1 to temperature T_2 through an isochoric process; (Given : Symbol have their usual meaning). Then choose the correct option(s)
- (A) $\Delta Q = \frac{3}{2}nR(T_2 - T_1)$ (B) $\Delta U = \frac{3}{2}nR(T_2 - T_1)$
 (C) $\Delta Q = \frac{15}{7}(nR)(T_2 - T_1)$ (D) $W = 0$

Space for Rough Work

9. An ideal gas is undergoing a cyclic process $A \rightarrow B \rightarrow C$ as shown in the figure. Choose the incorrect option(s) :



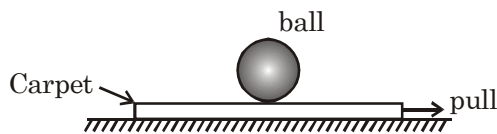
- (A) $P_A < P_B$ (B) $P_B = P_D$ (C) $P_D > P_C$ (D) $P_B > P_D$
10. A conducting rod of uniform cross section is kept at constant temperature (T_0) at one end, its another end is connected to a block of mass m , specific heat S at temperature $T < T_0$. Specific heat capacity of rod is negligible. Select the correct option(s) :



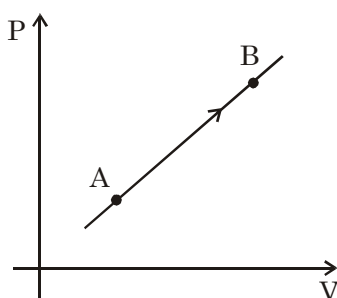
- (A) Rate of change of temperature of block is decreasing with time.
 (B) Energy is flowing at constant rate from end P to Q.
 (C) Temperature gradient at any instant is same at every point along the rod.
 (D) Rate of energy flow is decreasing with time.

Space for Rough Work

11. A ball lies on a carpet. Carpet is kept on smooth horizontal floor. Carpet is pulled as shown in figure. There is no slipping between ball & carpet.



- (A) Acceleration of center of mass of ball is less than acceleration of carpet.
 (B) Static friction force is acting between ball & carpet.
 (C) Ball is rotating anticlockwise
 (D) Work done by friction on ball + carpet system is non-zero.
12. For an ideal gas graph between pressure and volume is given then from A to B :- (Symbol have their usual meaning)



- (A) $W > 0$ (B) $Q > 0$ (C) $W < 0$ (D) $\Delta U > 0$

Space for Rough Work

SECTION-II : (Maximum Marks: 18)

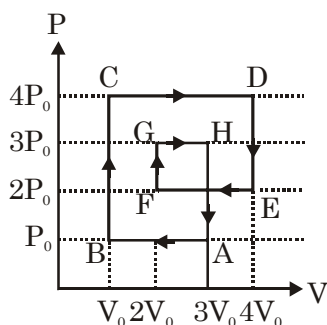
- This section contains **SIX** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33 , -0.30 , 30.27, -127.30 , if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.
For Example : If answer is -77.25 , 5.2 then fill the bubbles as follows.

										+	
1	1	1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9	9	9

										-	
1	1	1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9	9	9

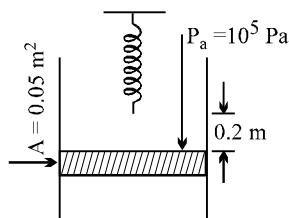
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. The work done by gas in process ABCDEFGHA is $2NP_0V_0$. Find the value of N.



Space for Rough Work

2. A cylinder fitted with a spring loaded piston shown in figure contains 0.01 m^3 gas at a pressure of 10^5 Pa . The cross-sectional area of the piston is 0.05 m^2 . Initially the spring does not touch the piston but atmospheric pressure of 10^5 Pa acts on the piston. The gas is slowly heated till the volume is increased to three times the original value. If the force constant of the spring is 200 kN/m and if workdone by the gas (in kilo Joule) is W . Find $\frac{W}{4}$.

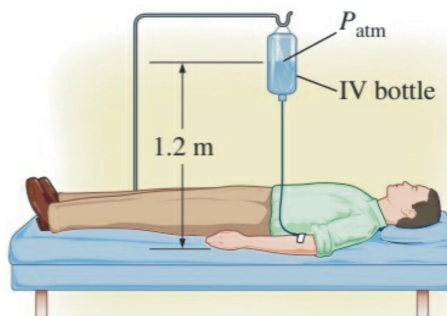


3. A mass $m_A = 2.0\text{ kg}$, moving with velocity $\vec{v}_A = (4.0\hat{i} + 4.5\hat{j} - 2.0\hat{k})\text{ m/s}$, collides with mass $m_B = 3.0\text{ kg}$, which is initially at rest. Immediately after the collision, mass m_A is observed travelling at velocity $\vec{v}_A = (-2.0\hat{i} - 2.0\hat{k})\text{ m/s}$. Find the speed (in m/s) of mass m_B after the collision. Assume no outside force acts on the two masses during the collision.
4. A glass flask contains some mercury at room temperature. It is found that at different temperatures the volume of air inside the flask remains the same. If the volume of mercury in the flask is 300 cm^3 , then find the volume of the flask. If your answer is N (in cm^3), fill value of $\frac{N}{250}$.

[Given that coefficient of volume expansion of mercury and coefficient of linear expansion of glass are $1.8 \times 10^{-4} (\text{ }^\circ\text{C})^{-1}$ and $9 \times 10^{-6} (\text{ }^\circ\text{C})^{-1}$ respectively]

Space for Rough Work

5. A closed container contains diatomic gas at temp T_1 (300 K), pressure P_1 . Gas is now heated to temperature T_2 (600 K) such that 20% of molecules are broken into atoms. Find ratio of final pressure (P_2) to initial pressure (P_1) $\frac{P_2}{P_1}$.
6. Intravenous infusion usually driven by gravity by hanging the fluid bottle at sufficient height to counteract the blood pressure in the vein and to force the fluid into the body (see figure). Higher the bottle is raised, higher is the rate of flow of fluid. It is observed that fluid and blood pressure balance each other when bottle 1.2 m above the arm level. Density of fluid is 1220 kg/m^3 . Find gauge pressure of blood in (kPa). ($g = 10 \text{ m/s}^2$)



Space for Rough Work

PART-2 : CHEMISTRY**SECTION-I(i) : (Maximum Marks : 12)**

- This section contains **FOUR** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
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Negative Marks : -1 In all other cases

-
1. The **CORRECT** set of statement(s) among the following.
- (i) It requires higher amount of energy to break one mole N_2 as compared to break one mole O_2
 - (ii) γ - SO_3 contains only $p\pi-d\pi$ type of covalent bonds
 - (iii) Bond energy of Br_2 is greater than F_2
 - (iv) Bond length of F_2 is greater than that of I_2
- (A) (i), (ii), (iii) (B) (ii), (iii) (C) (i), (ii) (D) (i), (iii)
2. $CO(g) + 2H_2(g) \rightleftharpoons CH_3OH(g)$; $\Delta H < 0$
- $K_c = 14.5$ at 483K, A mixture of $CO(g)$ & $H_2(g)$ is pumped into a previously evacuated 2.0 litre reaction vessel. The total pressure of the reaction system is 1.2 atm at equilibrium. What will be total pressure of the system if the volume of the reaction vessel is reduced to 1.0 litre at constant temperature -
- (A) Less than 1.2 atm
 - (B) Greater than 1.2 atm but less than 2.4 atm
 - (C) 2.4 atm
 - (D) Greater than 2.4 atm

Space for Rough Work

3. Only certain transitions are allowed from one energy level to another, in one electron species, the change in ' l ' of an allowed transition is ± 1 for example a '3p' electron can move to a 2s orbital but not to 2p. Thus in a UV series where ' $n_{\text{final}} = 1$ ' allowed transition can start from a 'p' orbital ($l = 1$) of $n = 2$ or higher, but not from a s ($l = 0$) or d ($l = 2$) of $n = 2$ or higher. From what orbital do each of the allowed transition start for the four emission lines in the visible spectrum ($n_{\text{final}} = 2$)

(i) 2 p to 2s

(iv) 4s $\rightarrow$ 2p

(ii) 3s to 2p

(v) 3p $\rightarrow$ 2s

(iii) 3d to 2p

(vi) 5s $\rightarrow$ 2s

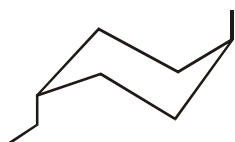
(A) only (ii), (iii), (iv) & (v)

(B) only (i), (ii), (iii), (v)

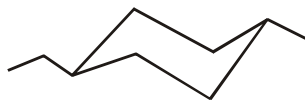
(C) only (i), (vi), (iii), (iv)

(D) only (i), (iii), (iv), (v)

4. Which of the following represent correct stability order of



(I)



(II)



(III)



(IV)

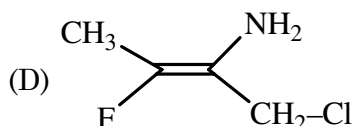
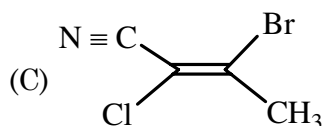
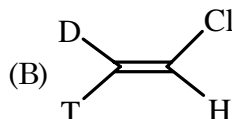
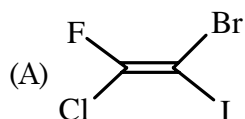
- (A) II > I > IV > III (B) II > III > IV > I (C) II > IV > III > I (D) II > IV > I > III

Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 32)

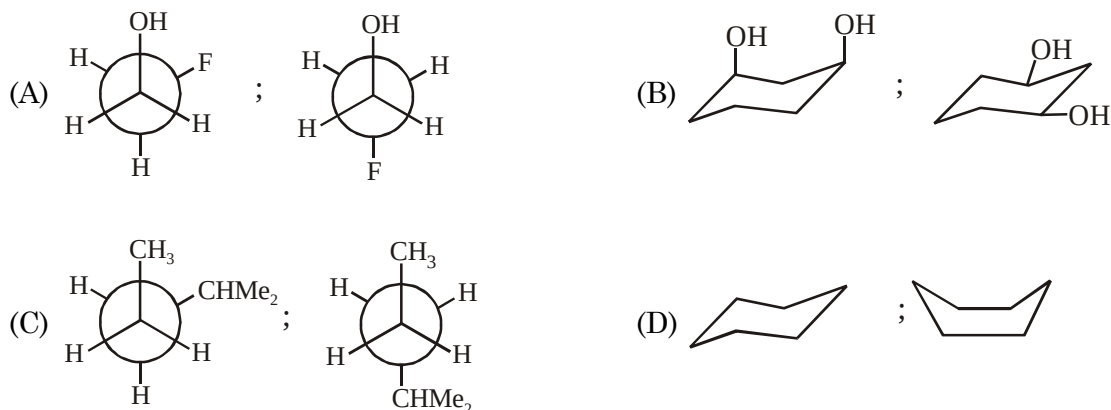
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5. Identify 'E' configuration ?



Space for Rough Work

6. In which case 2nd structure is less stable than 1st structure.



7. Which of the following Compound(s) are soluble in NaOH

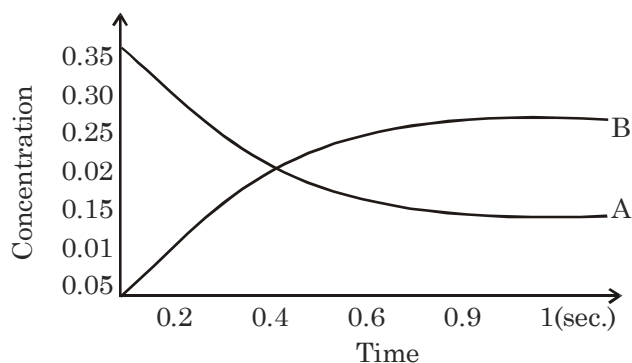


8. In 2.6 kg of $\text{FeSO}_4 \cdot 6\text{H}_2\text{O}$ (At. wt. of Fe = 56)

- (A) Number of atoms of O are $100 \times N_A$
 (B) Moles of H-atoms are $120 \times N_A$
 (C) Molecules of water of crystallisation are $60 \times N_A$
 (D) Moles of e^- present in SO_4^{2-} are 500

Space for Rough Work

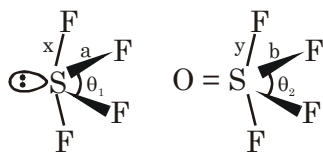
9. Above graph shows change in concentration of A & B for $A \rightleftharpoons B$ then which of the following are FALSE regarding the progress of reaction -



- (A) If the reaction is observed after 10 minutes the concentration of A & B would be equal
 (B) After 1 sec reaction has gone to completion
 (C) If 'A' is added at 2 min then equilibrium would adjust to consume 'A' and produce a new ratio of $\frac{[B]}{[A]}$.
 (D) If B is added at 3 min then equilibrium would adjust to consume 'B' and produce a new ratio of $\frac{[A]}{[B]}$.

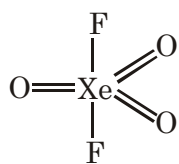
Space for Rough Work

10. Consider the following two molecules.



Which of the following order(s) is/are **INCORRECT**? (x, y, a & b are bond lengths, θ_1 & θ_2 are bond angles)

- (A) $x < y$ (B) $\theta_1 < \theta_2$ (C) $a < b$ (D) $a > b$
11. Consider the molecule



Which is/are **CORRECT** for the above molecule.

- (A) The orbital used by Xe to form pi-bond will definitely have two nodal planes
 (B) The charged comb will attract the above molecule when brought near it.
 (C) Exactly two obtuse bond angles are present
 (D) The molecule will be non planar
12. Which of the following order(s) is/are **INCORRECT**?
- (A) $F > Cl$ (Electronegativity & E.A.)
 (B) $H^- > Cl^-$ (Ionic size)
 (C) $Li^+ > Mg^{2+}$ (Ionic size & I.E.)
 (D) $S > O$ (Atomic size & E.A.)

Space for Rough Work

SECTION-II : (Maximum Marks: 18)

- This section contains **SIX** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darken the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	●	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

-									
●	●	●	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If ONLY the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. Consider the following elements.

Fe, Mg, F, Na, Cu, Zn, Rb, Mo, Br, Ba, Cl, K, Cd, Sr, Li

If x → Elements occupying descending positions in Lothar-Meyer curve

y → Elements occupying ascending positions in Lothar-Meyer curve

z → Elements lying in Broad Minima of Lothar-Meyer curve

p → Elements occupying peak positions in Lothar-Meyer curve

then the value of $\frac{(p+z)}{(x+y)}$ is

2. Borax is $\text{Na}_2[\text{B}_4\text{O}_5(\text{OH})_4] \cdot x\text{H}_2\text{O}$

Tetrapolyphosphoric acid is $\text{H}_{(2+y)}\text{P}_4\text{O}_{13}$

Basicity of pyrophosphorus acid is z,

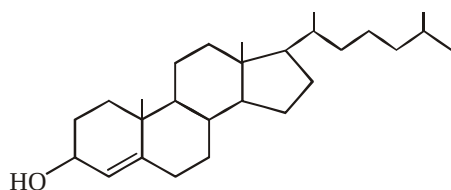
then the value of $\frac{(x+y+z)}{2}$ is

Space for Rough Work

3. Root mean square speed of an unknown gas at 727°C is 10^5 cm/second. Calculate molar mass of unknown gas (in gram/mole) [Take $R = \frac{25}{3}$ J/mole-K].
4. A solution of $\text{Na}_2\text{S}_2\text{O}_3$ is standardized iodometrically against 125.25 mg of KBrO_3 where BrO_3^- changes to Br^- . This process requires 45 mL of the $\text{Na}_2\text{S}_2\text{O}_3$ solution. What is the molarity of the $\text{Na}_2\text{S}_2\text{O}_3$? [Mw. of $\text{KBrO}_3 = 167$]

Space for Rough Work

5. Calculate total number of chiral centre present in following compound



6. How many compounds are theoretically possible for formula C_3H_6O (excluding stereoisomers)?

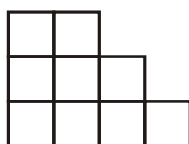
Space for Rough Work

PART-3 : MATHEMATICS

SECTION-I(i) : (Maximum Marks : 12)

- This section contains **FOUR** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

1. Persons A,B,C,D & E are to be seated on a circular table. Number of ways they can be seated so that exactly two from A,B & C are adjacent is equal to -
 (A) 6 (B) 8 (C) 12 (D) 24
2. The number of ways in which 5 X's can be placed in the squares of the figure so that no row remains empty is :



- (A) 97 (B) 98 (C) 100 (D) 126
3. The eccentric angle of a point on the ellipse $9x^2 + 4y^2 = 36$ at a distance $\frac{\sqrt{31}}{2}$ units from the centre of the ellipse is ' θ ', then the value of $|12\cos\theta|$ is -
 (A) 2 (B) 4 (C) 6 (D) 8
 4. The line $2x - y = 3$ is rotated through an angle $\frac{\pi}{4}$ in anticlockwise direction about the point $(2, 1)$, then equation of line in its new position is
 (A) $3x + y = 7$ (B) $3x - y = 7$ (C) $3x + y = 5$ (D) $3x - y = 5$

Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 32)

- This section contains **EIGHT** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -1 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

-
5. If point P lies on the curve $y^2 = 8x$ then locus of mid point of line segment PA is $(2y - 1)^2 = \lambda(x - 1)$ where A(2, 1). Then λ is divisible by
 (A) 2 (B) 4 (C) 6 (D) 8
6. Consider a photo session of 3 teachers, 6 boys and 5 girls in a line such that no two teachers are together, boys stand in increasing order of their height and girls in decreasing order of their height. If number of such arrangements is $k \cdot {}^{22}C_5$, then 'k' is divisible by (Assume all students to be of different heights) -
 (A) 8 (B) 10 (C) 12 (D) 15
7. From a point on the line $x - y + 2 = 0$ tangents are drawn to the hyperbola $\frac{x^2}{6} - \frac{y^2}{2} = 1$ such that the chord of contact passes through a fixed point (λ, μ) . Then $\frac{\lambda}{\mu}$ is less than or equal to -
 (A) 2 (B) 3 (C) 4 (D) 5
-

Space for Rough Work

8. If the difference of number of diagonals in $(n + 1)$ sided polygon & n sided polygon is 9, then -
 (A) n sided polygon is decagon
 (B) if this n quantities are divided into four groups such that number of elements in group are in A.P. with common difference 1, then number of ways are 12600.
 (C) the value of $\frac{1}{1.2} + \frac{1}{2.3} + \dots + \frac{1}{(n-1)n}$ is greater than 0.8
 (D) if ${}^nC_r = {}^nC_{r+4}$, then $r = 5$
9. If $x + 2 = 0$ and $y = 1$ are the equation of asymptotes of rectangular hyperbola passing through $(1,0)$. Then which of the following is(are) not the equation(s) of hyperbola -
 (A) $xy - x + 2y + 1 = 0$ (B) $xy - x - 2y + 1 = 0$
 (C) $xy - x - 2y - 1 = 0$ (D) $xy + 2x - y + 1 = 0$
10. Consider the parabola $y^2 = 12x$. Then
 (A) line $y = 3x + 1$ is a tangent to the parabola.
 (B) line $y = 2x - 36$ is a normal to the parabola.
 (C) line $2x + y = 6$ is a chord of contact for some point on directrix of the parabola.
 (D) line $x - 2y + 12 = 0$ subtends a right angle at the vertex of the parabola.
11. Pair of tangents are drawn from the point $P(9, 6)$ to the circle $(x - 5)^2 + (y - 2)^2 = 16$, then which of the following is/are true
 (A) one tangent is parallel to x-axis (B) one tangent is parallel to y-axis
 (C) angle between tangent is 45° (D) angle between tangent is 90°
12. For the hyperbola $16x^2 - 25y^2 - 32x + 50y - 409 = 0$ which is/are true -
 (A) foci are $(\sqrt{41} + 1, 1)$ and $(-\sqrt{41} + 1, 1)$ (B) eccentricity is $\frac{\sqrt{41}}{5}$
 (C) one of the directrix is $x = \frac{-25}{\sqrt{41}} + 1$ (D) length of latusrectum $= \frac{16}{5}$

Space for Rough Work

SECTION-II : (Maximum Marks: 18)

- This section contains **SIX** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darken the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	●	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	●	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	●	●	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

-									
●	●	●	0	0	●	●	●	●	●
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	●	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	●	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If ONLY the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. Consider a family of lines $(2k + 1)x + (3k - 2)y + (3 - 8k) = 0$ where $k \in \mathbb{R}$. The minimum area of the triangle formed by a line of given family with positive semi axes is equal to
2. From the centre C of the hyperbola $\frac{x^2}{16} - \frac{y^2}{9} = 1$, perpendicular CN is drawn on any tangent to it at the point P in the first quadrant. If A denotes the maximum area of the triangle CPN, then the value of (4A) is

Space for Rough Work

3. Through the vertex O of parabola $y^2 = 4x$, chords OP and OQ are drawn at right angles to one another, where P and Q are point on the parabola. If the locus of middle point of PQ is $y^2 = 2(x - \ell)$, then value of ℓ is
4. The co-ordinates of a point P(p, q) on the line $4x - y = 10$ such that $|PA - PB|$ is maximum where A is (2, 3) and B is (1, -2), then the absolute value of $\left(\frac{q}{p}\right)$ is

Space for Rough Work

5. Maximum distance between two points lying on the curve $4x^2 + 9y^2 + 8x - 36y + 4 = 0$ is
6. A light ray emerging from the point source placed at $P(1, 3)$ is reflected at a point Q in the axis of x . If the reflected ray passes through the point $R(6, 7)$, then the area of ΔPQR is

Space for Rough Work

Space for Rough Work

QUESTION PAPER FORMAT AND MARKING SCHEME :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

17. Each part has two sections as detailed in the following table.

Section	Que. Type	No. of Que.	Category-wise Marks for Each Question				Maximum Marks of the section
			Full Marks	Partial Marks	Zero Marks	Negative Marks	
I(i)	Single correct option	4	+3 If only the bubble corresponding to the correct option is darkened	—	0 If none of the bubbles is darkened	–1 In all other cases	12
I(ii)	One or more correct option(s)	8	+4 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened	+1 For darkening a bubble corresponding to each correct option, provided NO incorrect option darkened	0 If none of the bubbles is darkened	–1 In all other cases	32
II	Numerical Value Type (Up to second decimal place)	6	+3 If only the bubble corresponding to correct answer is darkened	—	0 In all other cases	—	18

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same.

Signature of the Invigilator